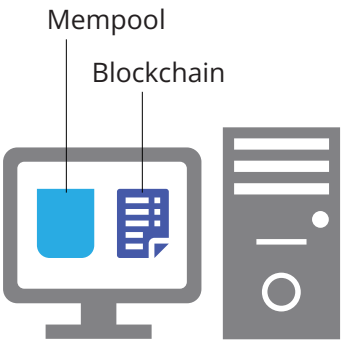
The **nodes** in the ***Bitcoin Network*** **are like puzzle checkers.** They have to verify that all the pieces fit together correctly. They make sure that Jim both owns the ***bitcoin*** and authorized the transaction.

Once a majority of the nodes are in consensus that the puzzle is solved correctly, the transaction is considered legitimate and is added to a **queue**.

This queue of pending transactions is called the "**mempool**."

The **mempool** is like a waiting area for puzzles that have been solved correctly, but haven't been added (chained) to the permanent puzzle (***blockchain***) yet. It is located in a different section of a node's memory storage compared to the *blockchain*, which permanently records confirmed transactions.

A Node in the Bitcoin Network



Once transactions are verified, they must be recorded permanently on the ***blockchain***. A group of **nodes** called "**miners**" compete to be the first to add them to the ***blockchain*** in order to receive a reward.

Meet the superheroes of ***Bitcoin***: **miners**! These special computers use their super-powered software to check that nobody is **double-spending**, stealing, or accidentally sending funds they don't have, and they make sure all the other miners are doing the same.

- **Miners** are like puzzle curators, they select which puzzles from the queue to add to the final puzzle exhibit, this process ensures that the same bitcoin can't be spent twice by the same person and that transactions are processed quickly.

Miners keep a copy of the *blockchain*, and they check each transaction against the blockchain to **confirm** that the same ***bitcoin*** haven't been spent before. Only legitimate transactions that meet certain criteria, such as having the correct digital signature and enough funds, are added to the *blockchain* by the miners.

Once on the ***blockchain***, the transactions included on the block are considered complete and irreversible. The exchange of ***bitcoin*** from one address to another is settled.